BIOT 6900 Module 5

Nicholas Adrogue

Gene: CYP2C19

Thresholds and Calibration

I used a 0.5 frequency ceiling as my threshold for the analysis. CYP2C19 is a pharmacogene which activates the antiplatelet prodrug clopidogrel. Focusing on the rare alleles of the gene therefore would not make sense, given that the alleles that affect the drug response are not rare. For example, *2 and *17 are two major alleles in the drug response, and they have AFs of 0.168 and 0.200, respectively[1].

The 0.5 frequency ceiling removed 2 of the 80 variants, rs17885098 and rs3758581. Both variants are benign and both are actually the majority allele. This is by design, given that the alleles that survived the filter are those below a frequency of 0.5. The former is a synonymous variant and the latter a missense variant. Neither allele is involved in drug response.

Using a 0.001 AF ceiling removed 19 variants, with 13 of them being involved in drug response, including *2 (rs4244285), *3 (rs4986893), and *4 (rs28399504). Some of them are specifically listed in AMP tier 1 as highly recommended alleles to study for pharmacogenetic testing, such as *2 and *3[2]. Therefore, including only rare alleles would exclude important data.

I calibrated CADD as the predictor, and pooled with DPYD. I chose CADD since there were non-missense classes of variants being looked at, and REVEL only looks at missense. After filtering, CYP2C19 had no P/LP, so pooling it with another gene was required. Using DPYD, there were 39 P/LP and 96 B/LB.

Table 1: Likelihood Ratios

CADD band	n	P/LP	B/LB	LR+	Evidence
0–10	65	0	65	0	Strong, Benign

10–20	24	0	24	0	Strong, Benign
20–25	11	5	6	2.05	No Evidence
25–30	4	3	1	7.38	Moderate, Pathogenic
30–50	31	31	0	inf	Strong, Pathogenic

The results from this calibration, therefore, are based much more on DPYD than CYP2C19. Furthermore, the LR+ of zero or infinity are caused simply from a lack of P/LP or B/LB in the given bands, respectively. This results in extremes that are too confident than the sample size can back up. The 25-30 band only rests on 4 variants, meaning that the sample size is too small, and while the 20-25 band rests on 11, the LR+ is a region of not indicating enough evidence either way[6].

Coverage

REVEL and AlphaMissense only see missense variants, leading to both missing 7 of the 9 variant types that are seen in CYP2C19, including those that include *2, which is synonymous, *17, which is upstream, and *3, which is stop-gained. One completely unscored variant was rs72558186. Evidence that could classify it would be data in RNA splicing.

The Atlas scored 70 of 78. The top hits were rs4986893 (*3), driven by protein termination, rs72558186, driven by merged splicing, and rs4244285 (*2), also driven by merged splicing. There were many driven by missense that had already been seen by the previous tools.

Discordant Variant: CYP2C19*2

CADD and Atlas disagreed on CYP2C19*2. CADD gave it a score of about 2.6. Atlas gave it an AVI of 22.3, given that it looked at how the RNA splicing changed. Given this high AVI, and the fact that it had no REVEL nor AlphaMissense score (because it is synonymous), it tripped the atlas_only flag. The relatively high AF of 0.168 and label of “other” also tripped the common_but_plp rule. Given the low CADD, it would have to be true that *2 is a fairly tolerated

variant, which tracks logically from the fact that it is a synonymous variant. However, the variant causes a splice site that was not there before[5], which accounts for the high AVI from Atlas.

Limitations and Future Work

The most interesting variant was *2. There were things this analysis could not cover. For example, the zygosity and phase. For this gene, the phenotype depends on both alleles in the genotype, leading to different expressions with different sets of alleles. The *2 variant participates in multiple of these phenotypes, including intermediate metabolizers, likely poor metabolizers, and poor CYP2C19 metabolizers[3].

Another missing area is the phenoconversion given the effect that taking other medicines can have on the CYP2C19 metabolism. In a study[4], it was shown that fluvoxamine especially can have a negative effect on CYP2C19 activity, including with *2 carriers.

An experiment that would resolve this would be a study of heterozygous carriers using samples from the liver. This would show the expression of the gene given different combinations of alleles. If there were no differences in the splicing (which exons or introns are present in the mRNA), then the conclusion drawn from Atlas would be incorrect.

Sources

[1] Lee CR, Luzum JA, Sangkuhl K, et al. Clinical Pharmacogenetics Implementation Consortium Guideline for CYP2C19 Genotype and Clopidogrel Therapy: 2022 Update. Clin Pharmacol Ther. 2022;112(5):959-967. doi:10.1002/cpt.2526. PMID: 35034351; PMCID: PMC9287492.

[2] Pratt VM, Del Tredici AL, Hachad H, Ji Y, Kalman LV, Scott SA, Weck KE. Recommendations for Clinical CYP2C19 Genotyping Allele Selection: A Report of the Association for Molecular Pathology. J Mol Diagn. 2018;20(3):269-276. doi:10.1016/j.jmoldx.2018.01.011.

[3] Lima JJ, Thomas CD, Barbarino J, et al. CPIC Guideline for CYP2C19 and Proton Pump Inhibitor Dosing. Clin Pharmacol Ther. 2021;109(6):1417-1423. doi:10.1002/cpt.2015. PMID: 32770672; PMCID: PMC7868475.

[4] de Jong LM, Boussallami S, Sánchez-López E, et al. The impact of CYP2C19 genotype on phenoconversion by concomitant medication. Front Pharmacol. 2023;14:1201906. doi:10.3389/fphar.2023.1201906.

[5] de Morais SM, Wilkinson GR, Blaisdell J, Nakamura K, Meyer UA, Goldstein JA. The major genetic defect responsible for the polymorphism of S-mephenytoin metabolism in humans. J Biol Chem. 1994;269(22):15419-15422. PMID: 8195181.

[6] Tavtigian SV, Greenblatt MS, Harrison SM, et al. Modeling the ACMG/AMP variant classification guidelines as a Bayesian classification framework. Genet Med. 2018;20(9):1054-1060. doi:10.1038/gim.2017.210. PMID: 29300386.